

Exam Preparation

This exam preparation tutorial focuses on the Piaggio P180 aircraft. Several calculations performed during the previous tutorials during the term for the A320 will be repeated for this aircraft. This tutorial is intended to serve as a revision of some of the presented methodologies for the exam. However, it is not to be seen as exhaustive. Refer to the lecture notes for the theory.



Source: Wikimedia Commons

Calculation tasks

- 1) Which configurative features are visible on the three-side view and the illustration of each aircraft? What are the corresponding derived properties for the aircraft? List at least 6 for the Piaggio P180 moderate speed ($M \approx 0.6$), laminar flow executive transport aircraft.
- 2) Determine the lift coefficient slope of the horizontal stabilizer at $M = 0.5$ on aircraft level. Due to the vertical position, the local dynamic pressure is reduced by only 5% compared to the main wing and the effective angle of attack is reduced by 10%. If more than one method is applicable, use the most precise one.
- 3) Calculate the parasite drag area of the main wing during subsonic cruise flight at $h = 12000\text{ m}$ and $M = 0.6$. Assume laminar conditions prevail on 45% of the (camouflaged aluminum) wing chord and weight the equations for the coefficient of friction accordingly.
- 4) Locate the absolute location of the aerodynamic center of the horizontal tail during approach ($M = 0.3$), measured from the nose of the aircraft.
- 5) Estimate the mass of the fuselage in kilograms. The design dive speed is 400 kn .
- 6) An upgraded version of the aircraft shall use two turbofan engines in order to reach a nominal thrust to weight ratio of $T_0/w_{TO} = 0.4$. With the help of the sizing equations, estimate the length of the new nacelles. Cruise Mach number $M = 0.6$. Assume that the BPR is lower than 6.
- 7) Estimate the maximum lift coefficient increment when the (average) double-slotted trailing edge flaps are deflected by $\delta_{TE} = 20^\circ$. Assume that 80% of the wing area is influenced by the flaps, which have a chord of $c_{TE} = 0.2\text{ m}$.
- 8) Draw the limit curve for a take-off distance of $s_{acc} = 650\text{ m}$ at an altitude of 2000 m above mean sea level into a design chart (limit the number of supporting points of the resulting curve to three) and indicate the location of the design region. $C_{L,max,TO} = 2.2$, $C_{D0|tot} = 0.016$, $C = 1.15$, $C_{L,TO} = 1.66$, $\mu_f = 0.02$.

Assumptions

- Gravitational acceleration: $g = 9.81\text{ m/s}^2$

Dataset Piaggio P180

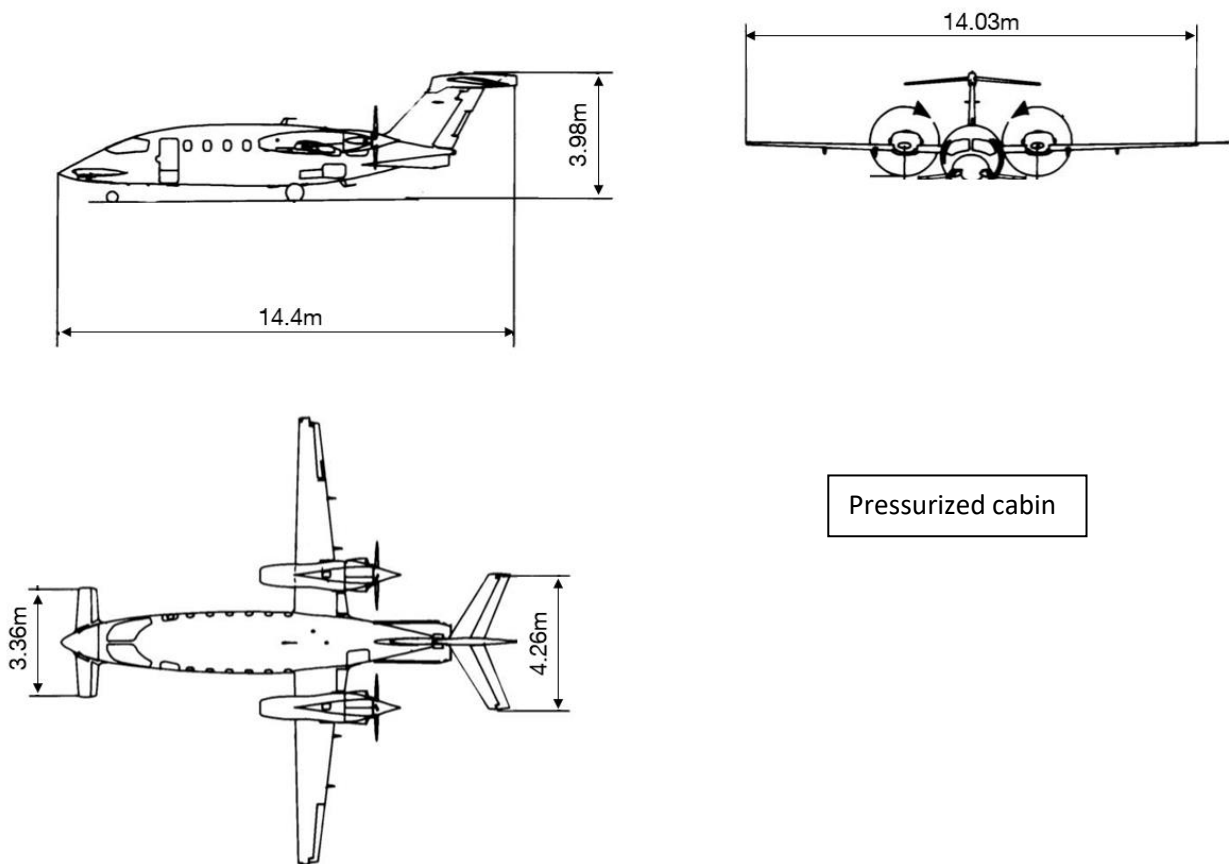
Geometric data

	Description	Symbol	Value	Unit
Wing	Area	S_{ref}	16	m^2
	Mean aerodynamic chord	c_{MAC}	1.284	m
	Average airfoil rel. thickness	$\left(\frac{t}{c}\right)_W$	14.5	%
	Airfoil type	NACA airfoil		
	Aspect ratio	AR_W	11.96	-
	Oswald factor at take-off	e_{TO}	0.80	-
	25% sweep angle	$\Lambda_{25\%,W}$	0.0194	rad
	% location of max. thickness	x_{max_t}	40	%
	Taper ratio	λ_W	0.371	-
	Distance from nose to wing apex	$x_{0,W}$	7.29	m
Horizontal tail	Area	S_{HT}	3.83	m^2
	25% sweep angle	$\Lambda_{25\%,HT}$	0.5145	rad
	Taper ratio	λ_{HT}	0.58	-
	Aspect ratio	AR_{HT}	4.57	-
	Average airfoil rel. thickness	$\left(\frac{t}{c}\right)_{HT}$	10.4	%
	Root chord	$c_{root,HT}$	1.10	m
	Distance from nose to horizontal tail apex	$x_{0,HT}$	12.07	m
Vertical tail	25% sweep angle	$\Lambda_{25\%,VT}$	0.8029	rad
	Aspect ratio	AR_{VT}	1.1675	-
	Taper ratio	λ_{VT}	0.52	-
	Distance from nose to vertical tail apex	$x_{0,VT}$	9.846	m
	Root chord	$c_{root,VT}$	2.53	m
Fuse-lage	Total length	l_{fus}	14.4	m
	Width	w_{fus}	1.85	m
	Height	h_{fus}	1.75	m

Mass data

	Take-off mass	m_{TO}	5239	kg
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Three view

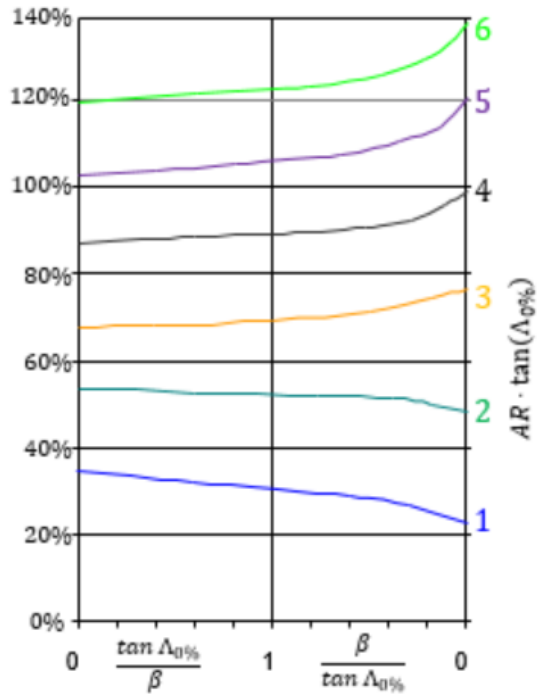


Unit Conversion Factors

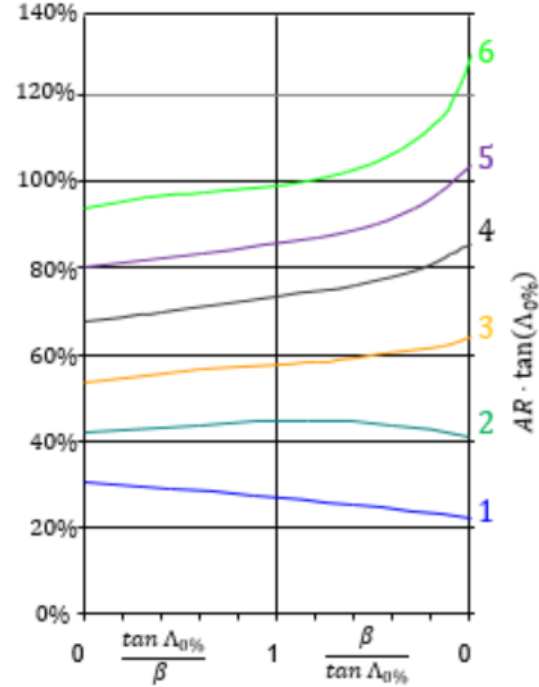
Length	1 m = 3.2808 ft
Area	1 m ² = 10.7639 ft ²
Mass	1 kg = 2.2046 lbm
Force	1 N = 0.2248 lbf
Velocity	1 m/s = 1.944 kn

Additional Data

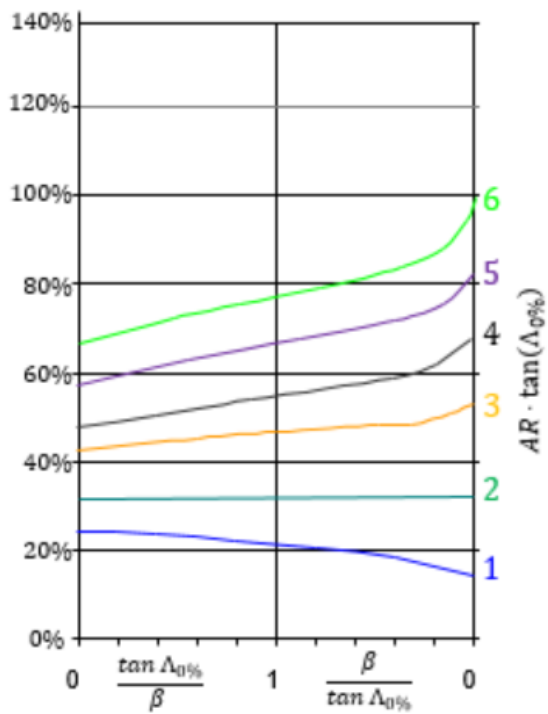
$\lambda = 0.5$



$\lambda = 0.25$

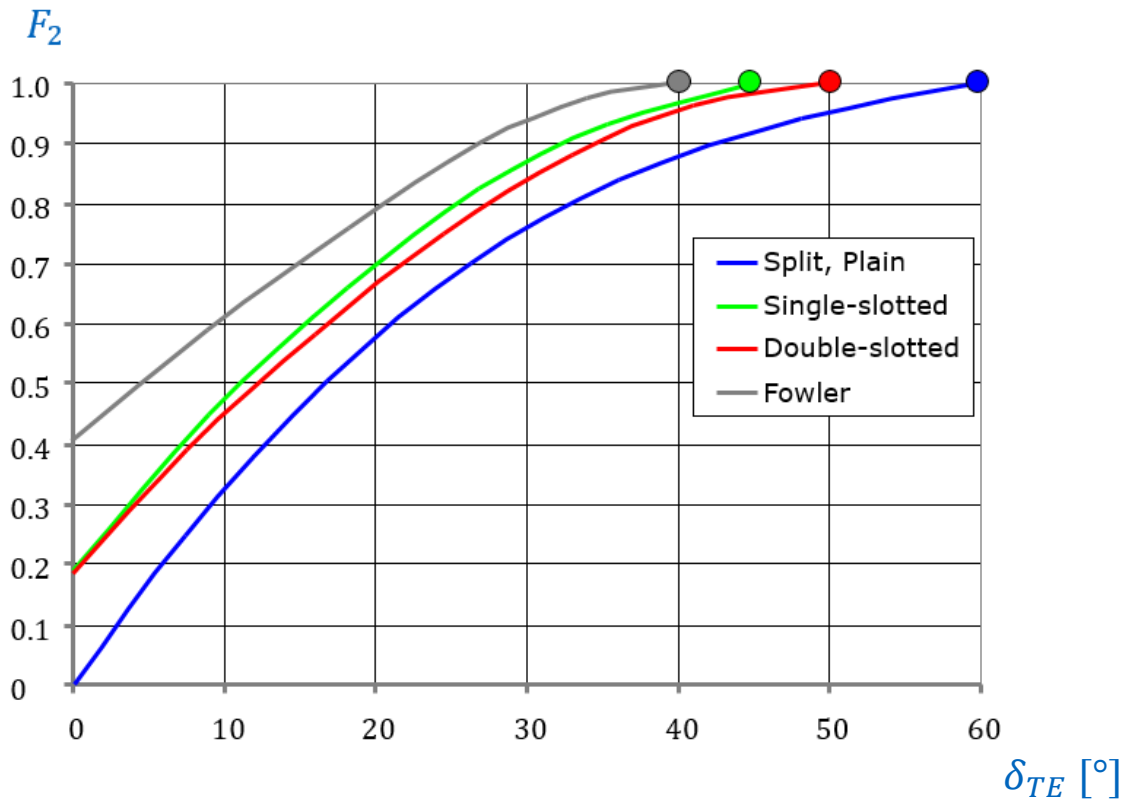
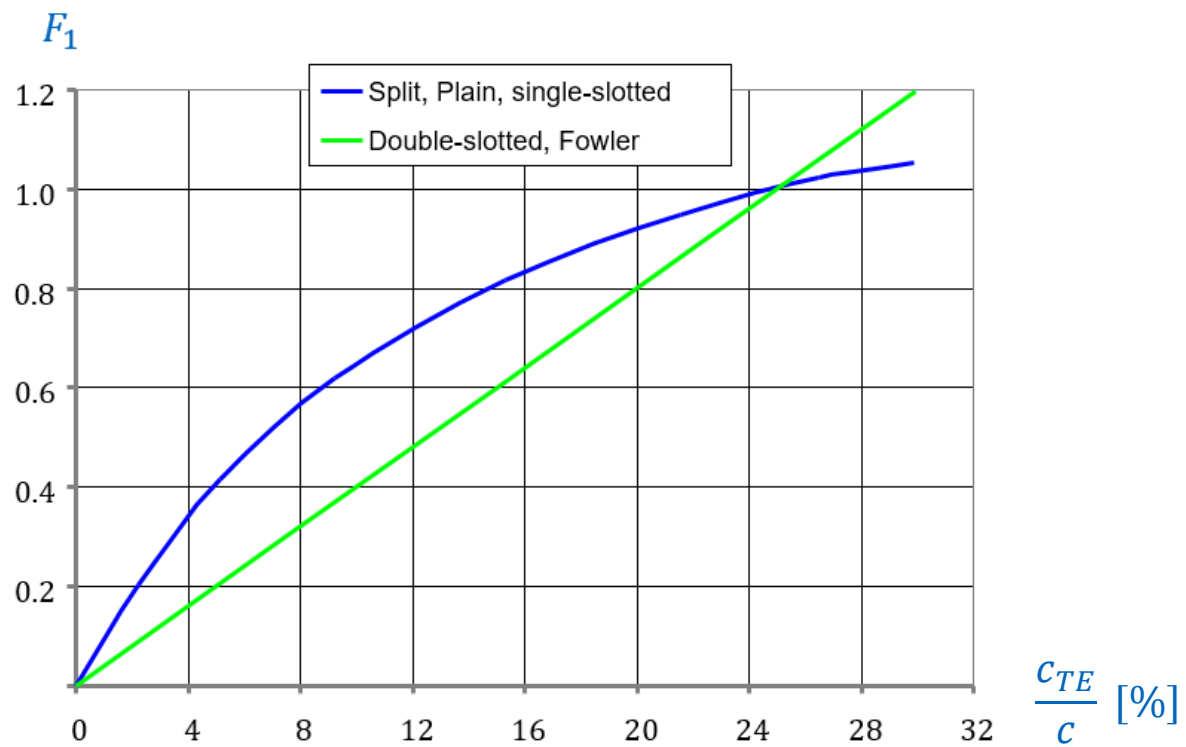


$\lambda = 0$

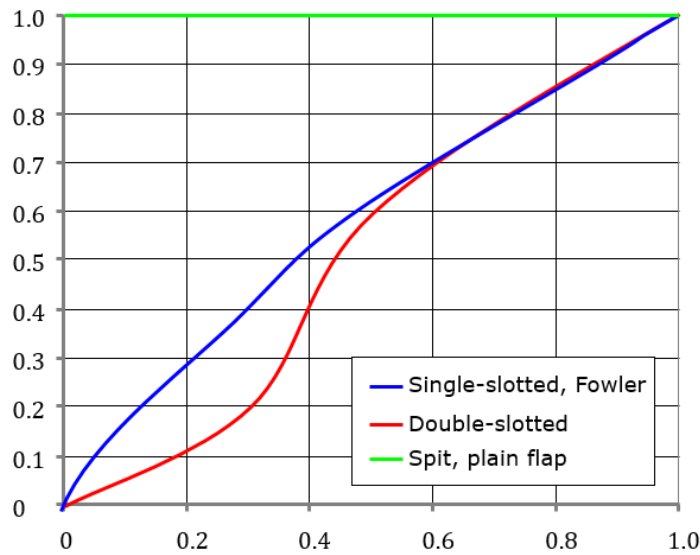


Surface	$k [m]$
Camouflaged aluminum	$10 \cdot 10^{-6}$
Flat coating	$6.3 \cdot 10^{-6}$
Unpolished sheet	$4.1 \cdot 10^{-6}$
Polished sheet	$1.5 \cdot 10^{-6}$
Flat CFRP	$0.5 \cdot 10^{-6}$

Trailing edge system lift increment – Roskam method



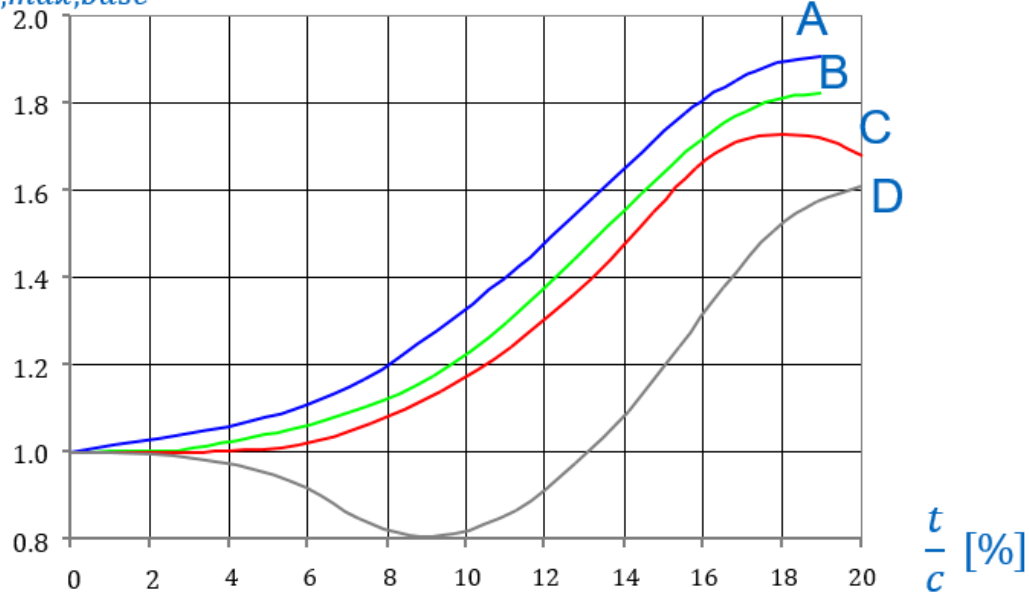
F_3



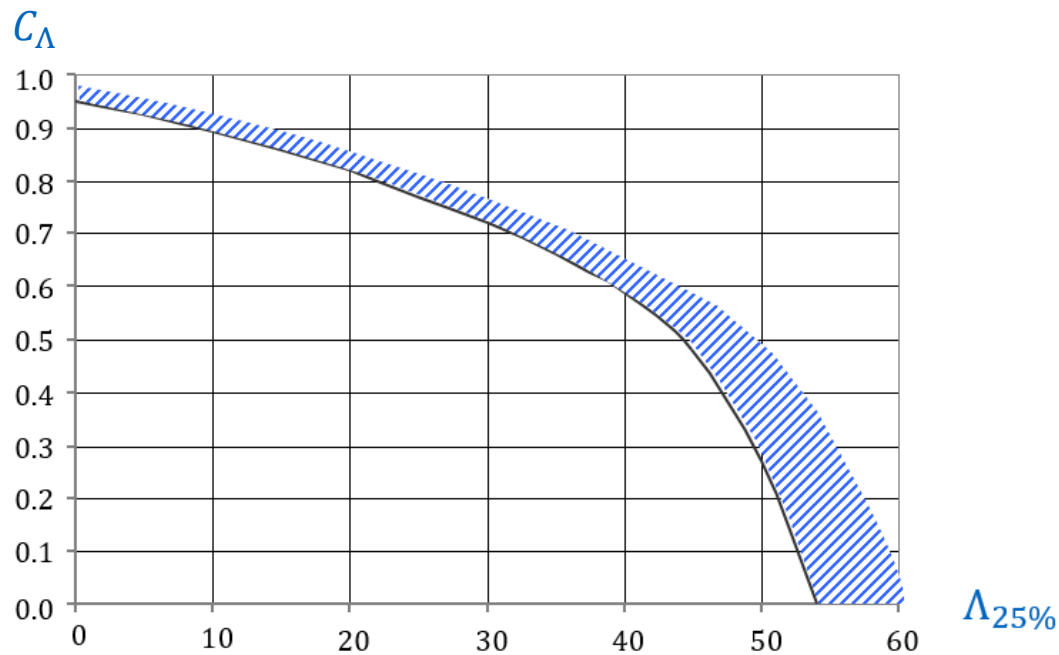
$\delta_{TE} / \delta_{ref} [^\circ]$

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$\Delta c_{l,max,base}$



	Flap type	Airfoil type
A	Optimum double-slotted	NACA
B	Average double-slotted	NACA
	Fowler	any
C	Double slotted	NACA 6 series
	Single slotted	any
D	Split / plain	any



Equations

$$K^2 = 1 + 0.75 \cdot \left(\frac{t}{c}\right)$$

$$\tan(\Lambda_{m\%}) = \tan(\Lambda_{n\%}) - \frac{4}{AR} \cdot \left[\frac{m-n}{100} \cdot \frac{1-\lambda}{1+\lambda} \right]$$

$$F_W = \left[1 + \frac{0.6}{(x/c)_m} \cdot \left(\frac{t}{c}\right)_W + 100 \cdot \left(\frac{t}{c}\right)_W^4 \right] \cdot [1.34 \cdot M^{0.18} \cos(\Lambda_m)^{0.28}]$$

$$Re_{CO,W} = 38 \cdot \left(\frac{l_{ref}}{k}\right)^{1.053} \text{ (subsonic); } Re_{CO} = 44 \cdot \left(\frac{l_{ref}}{k}\right)^{1.053} \cdot M_\infty^{1.16} \text{ (transsonic/supersonic)}$$

$$c_{f,lam} = \frac{1.328}{\sqrt{Re}}$$

$$c_{f,turb} = \frac{0.455}{(\log_{10} Re)^{2.58}} \cdot (1 + 0.144 \cdot M_\infty^2)^{-0.65}$$

$$S_{wet,W} = 2 \cdot S_{ref}$$

$$m_{fus} = 0.021 \cdot C_{fus} \cdot \sqrt{V_D \cdot \frac{d_{HT}}{w_{fus} + d_{fus}}} \cdot S_{fus}^{1.2} \text{ with } S_{fus} \approx l_{fus} \cdot \pi \cdot \left(\frac{w_{fus} + h_{fus}}{2} \right) \text{ \& } d_{HT} \approx x_{0,HT} -$$

$x_{0,W}$

with:

$$C_{fus} = 1 \cdot [\text{x1.04} \text{ x1.07} \text{ x1.08}]$$

- pressurized cabin (y/n)
- main landing gear attached to fuselage (y/n)
- rear-fuselage mounted engines (y/n)

$$l_{nac} = 0.49 \cdot \left(\frac{T_{unit}}{1000} \right)^{0.4} \cdot M^{0.2}$$

$$\Delta c_{l,max(TE)} = F_1 \cdot F_2 \cdot F_3 \cdot \Delta c_{l,max,base}$$

$$\Delta C_{L,max(TE)} = \Delta c_{l,max(TE)} \cdot \frac{S_{W,TE}}{S_{ref}} \cdot C_A$$

$$s_{acc} = \frac{1}{2} \cdot \bar{a} \cdot t^2 = \frac{V_{TO}^2}{2 \cdot \bar{a}}$$

$$\frac{\bar{w}}{g} \cdot \bar{a} = \bar{T} - \bar{D}$$

$$\bar{D} = \bar{D}_f + \bar{D}_{aer}$$

$$\bar{D}_f = \mu_f \cdot (w_{TO} - \bar{L}_{TO})$$

$$D_{aer} = D_{0|clean} + \Delta D_{flap} + \Delta D_{LG} + D_i$$

$$\bar{L}_{TO} = \frac{L_{TO(V=0)} + L_{TO(V=V_{TO})}}{2}$$

$$\bar{q} = \frac{q(V=0) + q(V=V_{TO})}{2}$$

$$C_{Di} = \frac{(C_{L_{TO}} - C_{L,0})^2}{\pi \cdot AR \cdot e} - \frac{C_{L,0}^2}{\pi \cdot AR \cdot e}$$

$$C_{L,0} \approx \frac{1}{3} \cdot C_{L,TO}$$

$$V_{TO} = C \cdot V_S$$